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Creative foraging and the explore–exploit trade-off in knowledge networks

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ABSTRACT

The scope of knowledge is constantly evolving, due to such factors as environmental changes, cultural evolution, and scientific discovery. Consequently, we are frequently confronted with gaps in our knowledge, compelling us to seek information from available sources. Sometimes the information we seek is easy to find; other times it has yet to be established by others, requiring us to creatively come up with an original perspective. Yet, little is known about how our foraging strategies change depending on the ease with which the information we seek is readily available. We investigated how the need to generate new ideas influences the rate at which individuals explore or exploit existing information. Participants ($N=138$) answered questions either fully answerable (low-creativity condition) or not fully answerable (high-creativity condition) with information they foraged for on Wikipedia. We created knowledge networks from the foraged information, wherein Wikipedia pages were nodes. The edges linked pairs of Wikipedia pages when they were visited by the participant either sequentially or within the same condition, and were weighted based on the semantic similarity between the pair of pages. This approach allowed us to measure exploration (jumping between disparate pages) and exploitation (viewing closely related pages). In the high-creativity condition, participants were more likely to trade-off between exploration (lower average edge weights) and exploitation (higher average clustering coefficients). This trade-off was associated with responses that were more novel, diverging further from the Wikipedia text, compared to less novel responses. These findings reveal how foraging strategies differ in creative versus non-creative contexts, and provide insight into the processes that underlie learning and scientific discovery.

1. Introduction

Human knowledge grows continuously as we strive to make sense of an ever-changing world, where environmental changes, cultural shifts, and scientific advancements generate an influx of new information. The sheer volume and dynamic nature of this new information makes it common for us to encounter something that we do not know. When this happens, we can choose to seek out or forage for the missing information by asking a friend or colleague, reading a book, or searching the internet for an answer (Hills et al., 2015; Pirolli, 2005). But, sometimes the answer for which we seek has yet to be articulated by others, leaving us to creatively come up with an answer based on (at times) loosely related information. This condition mimics what many scholars at the cutting edge of science experience when their questions cannot be solely answered by existing literature, causing them to formulate new perspectives. Conversely, cases where information can be found is reminiscent of a classroom setting, where a teacher poses a question that can be answered by looking through the course material. These

two contexts commonly occur throughout our personal and professional lives. Yet, little research has examined how these conditions shape our foraging behavior. In this study, we sought to uncover the foraging patterns associated with these two contexts: (1) low-creativity context — where the information sought can easily be found and (2) high-creativity context — where it is difficult to find relevant information, leading one to generate a new idea. We studied these behaviors using a naturalistic environment, where people are tasked with arriving at an answer using information they find on Wikipedia.

Extensive literature has examined the factors that motivate information search, including its connection to creativity (Kedrick et al., 2023; Koutstaal et al., 2022; Torunsky et al., 2025). Information search and creativity are often studied through semantic search, where creativity is thought to emerge when people retrieve concepts from memory that are far apart in semantic space (Kenett, 2018; Mednick, 1962; Volle, 2017). This work often treats creativity as an individual difference in memory structure: highly creative people are thought to have richer

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associative networks with more distant connections between concepts, while less creative people tend to rely on a smaller set of common associations (Kenett et al., 2014; Mednick, 1962). These memory structures of creative individuals have been associated with broader search processes, enabling access to more unconventional concepts or conceptual combinations (Kenett & Austerweil, 2016; Stella & Kenett, 2019).

In contrast, far less research asks how creativity shapes the way we search for information using external sources. Rather than defining creativity solely as an individual difference, we examine how creative contexts — situations in which people cannot rely on others' information — alter information-seeking strategies and search patterns. One way it might affect search strategies is by altering the frequency at which an individual exhibits exploration or exploitation. These distinct foraging patterns describe how an organism interacts with a patch of (for example) food or information (Hills et al., 2008, 2015). Exploitation can be thought of as honing in on a single patch, and depleting it of what it has to offer; whereas exploration captures the movement between patches.

While searching online, we can think of patches as topics or concepts that may be discussed within a single web page or across multiple web pages. During exploitation, a person may exhaust a patch by looking at all of the web pages that pertain to the same topic, or highly related content. In contrast, we can view exploration as jumping between loosely related web pages or patches of web pages. The choice to explore or exploit may depend on various factors: whether the current patch is depleted or the return from the current patch seems lower than what is perceived to be found in other patches (Hutchinson et al., 2008; Pirolli, 2005; Wilke et al., 2009a), whether the individual was primed to explore or exploit (Hills et al., 2008), or a general preference for one behavior over the other (Lydon-Staley et al., 2021). We extend these findings by testing whether the creativity required to formulate an answer influences the frequency with which a person explores or exploits.

While foraging, individuals draw connections between visited information, thereby forming networks of knowledge. We can use these knowledge networks to determine whether people were exploring, exploiting, or trading-off between the two. Lydon-Staley et al. (2021) constructed knowledge networks out of the foraging behavior exhibited by participants as they foraged through Wikipedia pages, in which Wikipedia pages were nodes that were linked if they were viewed by the same participant. They then used these networks of foraged-for information to articulate distinct foraging patterns. Specifically, their results suggest that some people's foraging strategy mimics hunters, wherein the hunter follows only one scent or topic, exploiting patches or clusters with the same or similar scent. While others are busybodies, who explore loosely connected topics without depleting a patch. These general characterizations map well onto the notions of exploration and exploitation. The hunters' behavior can be characterized as exploitation, wherein they exploit the information from highly related patches or topics. Busybodies can be thought of as explorers, who frequently jump between different patches of divergent topics. Associating graph-theoretic methods with distinct foraging patterns allows us to measure how people search for information.

Creativity may be associated with more exploratory behavior. Research suggests that exploration helps people arrive at creative solutions or novel ideas (Evans et al., 2021; Hass, 2017; Lee et al., 2021; Liu & Liang, 2020). For example, Hass (2017) suggests that when people search their memory for alternative ways to use common objects, they do not exploit clusters of similar uses. Rather, they use an exploratory approach, by responding with loosely associated ideas. A similar phenomenon may occur when foraging for information to create a new theory or explanation, wherein a person jumps around to disparate pages of semantically dissimilar topics forming loose networks of foraged-for information. In doing so, they may exhibit a generalist strategy, broadly sampling a wide range of loosely related information.

In contrast, when a problem does not require creativity or originality, people may exhibit higher levels of exploitation. That is, people may exploit a topic or a patch of web pages containing relevant information that has already been established as connected, with semantically similar content, thereby forming tighter knowledge networks of foraged information. The notions of exploration and exploitation in the learning literature distinguish *Learning how* from *Learning about* (see Giraldeau and Caraco (2000) Chapter 10, pages 271–281). Learning how is an exploitative use of knowledge, and it covers information that humans or other animals can use to imitate answers, or the procedures to generate answers without needing explicit knowledge of the relations between objects, causal mechanisms and events that would be needed to synthesize novel answers and solutions. In animals, observing when and where other animals are getting food can profoundly inhibit direct learning of food distribution, limiting both diet diversity and creating a situation called “Producer/Scrounger games”. In this context, some animals act as producers, generating knowledge independently, while others function as scroungers, exploiting the discoveries of others instead of investing effort in independent search or developing original strategies. Additional research suggests that exploitation is the best strategy for solving a problem when an individual can employ established principles (Benner & Tushman, 2001; Lavie & Rosenkopf, 2007). These findings suggest that exploitation may occur when the required information is well articulated; whereas it may be optimal to explore when the information is ill-defined.

Alternatively, when asked to be creative or come up with an original idea, people may trade-off between exploration and exploitation while foraging for information. That is, someone may start by exploiting one topic, before moving to or exploring a new, dissimilar topic that they thereby exploit. The combination of both exploitation (deeply understanding a topic by examining related information) and exploration (exposure to diverse ideas) may foster creative thinking or ideation, where knowledge from distinct topics is synthesized to create new ideas. This may lead to a search process that utilizes the benefits of both exploration and exploitation, and in turn form knowledge networks that exhibit clusters of semantically similar information that are semantically dissimilar from each other. Trading-off between exploration and exploitation has been shown to positively influence team performance (Benner & Tushman, 2001; Uotila et al., 2009) as well as reinforcement learning (Cohen et al., 2007; Hills et al., 2015; Sutton & Barto, 1998). It has also been the preferred strategy during creative-visual foraging tasks for both children and adults (Hart et al., 2022, 2017). For example, Hart et al. (2017) found that when asked to develop novel shape configurations, people exploited foraged-for configurations of a shape category before moving to or exploring a new category. These findings may extend to a semantic or informational context, where people forage for information.

To test how creativity impacts foraging behavior, we developed the Knowledge-Foraging Task, where people were presented with four questions and asked to answer the questions by searching through Wikipedia pages. The questions differed in the amount of creativity required to answer them: in the high-creativity condition, participants answered questions where it was difficult to find the answer on Wikipedia; in the low-creativity condition, participants answered questions where the answer could be located on multiple Wikipedia pages. Wikipedia provided an ecologically valid context, where participants could choose to exploit semantically similar information (by visiting related Wikipedia pages or clusters of pages) or explore semantically dissimilar information (by visiting diverse Wikipedia pages or jumping between clusters). In addition, our paradigm allowed participants to enter their answers in a textbox located on one side of the screen, enabling us to collect the text they generated while viewing the content of the Wikipedia pages.

To measure differences in exploration and exploitation, we created knowledge networks out of the foraged-for information, wherein Wikipedia pages were nodes. The edges linked pairs of Wikipedia

pages when they were visited by the participant either sequentially or within the same condition, and weighted based on the semantic similarity between pairs of pages. Creating knowledge networks out of the foraged-for information allowed us to characterize exploration as jumping between disparate pages (indicated by lower average edge weights and longer average path lengths) and exploitation as viewing highly related pages or pages belonging to the same topic (indicated by higher clustering coefficients). We conceptualize the choice to explore or exploit as occurring each time a participant selects the next page to visit. Participants may choose to exploit by visiting a page on a topic similar to those they have already visited, or they may choose to explore by moving to a semantically distant page that differs from their prior visits. In our analyses, we considered these behaviors as occurring across the entire task, examining patterns of exploration and exploitation throughout participants' search for answers.

Exploitation involves continuing within a cluster of related pages, by following links that are directly visible on the current page or by searching terms that are closely associated with the current content in memory. In this sense, exploitation relies on knowledge that is immediately available and requires little additional search effort. The best case is that exploitation yields steady progress toward relevant information with minimal cognitive cost. The worst case is that it produces a long, winding path that delays reaching relevant information, or that it leads to no relevant information at all. Exploration, by contrast, requires leaving that cluster to test more uncertain paths, often by visiting pages that are less immediately accessible. People may choose to explore when the current cluster seems exhausted, when progress toward the goal has slowed, or when they anticipate that a more distant cluster may contain promising new information. The best case is that exploration produces a rapid leap to previously inaccessible and relevant knowledge. The worst case is that it leads to wasted effort and dead ends. Because exploration is more cognitively demanding and less predictable than exploitation, participants must weigh its potential payoff against its risks.

We hypothesized that the foraging patterns could go one of two ways. First, we hypothesized that when tasked with generating creative responses (high-creativity condition), participants will exhibit more exploration as they jump between diverse ideas; alternatively, for cases where the participant can find relevant information (low-creativity condition), they will exploit what they found by honing in on similar clusters of pages (referred to as Hypothesis A). Second, we hypothesized that foragers tasked with arriving at creative responses (high-creativity condition) will trade-off more frequently between exploration and exploitation than those that do not need to create a new response (low-creativity condition; referred to as Hypothesis B). Rather than treating creativity as an individual difference, where creative responses are defined as those that deviate from other responses to the same question (Kenett, 2018; Silvia et al., 2008), our work defines creativity as deviations from the perspectives participants actively choose to engage with (e.g., the content on the Wikipedia pages).

Our results suggest that creative foraging is associated with trading-off between exploration and exploitation. Further analysis also revealed an association between the explore-exploit trade-off and more original responses, which deviated further from the content of the Wikipedia pages than less original responses. As a secondary analysis, we examined how knowledge networks related to individual differences in originality and found that more original responses (relative to others who saw the same stimuli) were associated with exploratory behaviors. Finally, we investigated whether the Marginal Value Theorem (MVT) could explain the participants' foraging behavior. We found that MVT did not differ significantly between conditions and that both conditions deviated significantly from the theorem's predictions.

2. Method

2.1. Participants

A total of 171 participants took part in the experiment. We determined this sample size based on a power analysis, which indicated that 157 participants would achieve a statistical power of 0.8 for detecting a small effect size of 0.2 using a paired t-test. We based our effect size on the small effect size reported in Lydon-Staley et al. (2021) who used similar network metrics (e.g., average clustering coefficients). The results for 33 participants were omitted: 20 because they missed both attention checks that asked about the instructions for each condition, 12 were excluded for poor task engagement (i.e., 11 did not click on Wikipedia pages during the task and 1 clicked only on a single page across all stimuli), and one was excluded after reporting to the experimenter that they had used their phone to look up information during the study. Therefore, the final number of participants was 138 (65 females, M age = 26.94 years, SD = 7.95). A post-hoc power analysis of our primary findings — focusing on the least extreme result from a paired t-test comparing average edge weights between conditions — indicated that the analysis was not underpowered, even with the reduced sample size (power = 0.99), based on a p -value of 0.001 and a Cohen's d of -0.49. Note that, from those 138 participants, 79 participants answered both instruction attention checks correctly, and 59 participants answered one of the two correctly while also passing at least half of all attention checks (i.e., two out of four), indicating that they were generally paying attention. All participants were required to be at least 18 years old and to have learned English by the age of 6 years.

61 of the participants were undergraduates from a large Midwestern university, and awarded extra course credit points in exchange for their participation (50 females, M age = 19.92 years, SD = 1.28). 77 of the participants participated through Prolific (an online platform that administers experiments to a pre-screened set of international participants), and were awarded \$12.00 per hour in exchange for their participation (15 females, M age = 32.42 years, SD = 6.53). The undergraduate and Prolific participant pools did not produce statistically different search behaviors across key metrics (average edge weights: t = -0.49, p = 0.63; average path length: t = -1.22, p = 0.22; average clustering coefficient: t = -1.28, p = 0.20). Our research protocol was approved by the Institutional Review Board (IRB).

2.2. Procedure

After providing informed consent, the participants took the Knowledge-Foraging Task, where they were asked to answer four questions based on information they found while browsing Wikipedia, and given a total of 15 min per question to develop an answer. Overall, each condition took 30 min, with a five-minute break in between conditions. After they finished the task, participants were thanked, debriefed, and compensated.

2.3. Knowledge-foraging task

For the Knowledge-Foraging Task, participants were given four questions, two per condition (See Fig. 1). For the low-creativity condition, participants were presented with two questions in which the answers were available on the Wikipedia pages. For the high-creativity condition, participants were presented with two questions that were difficult to answer using information directly found on Wikipedia, and required them to creatively come up with an original response. The task began with a practice stimulus that allowed participants to familiarize themselves with the experimental environment, which included a split screen that the participants could move between. On the left side of the screen was the prompt and a textbox where the participant typed their responses, which we administered using jsPsych (de Leeuw

et al., 2023). On the right side of the screen was a cloned version of Wikipedia, where participants could forage for new information.

After finishing the practice stimulus, participants were presented with a block of two questions from the high-creativity condition and a block of two questions from the low-creativity condition. The order of conditions was counterbalanced. To diversify the type of questions participants were asked, we devised two unique sets of stimuli for the high- and low-creativity conditions. Thus, we administered four versions of the task that varied based on the order of conditions and the set of stimuli: (1) high-creativity condition first, stimulus set A; (2) high-creativity condition first, stimulus set B; (3) low-creativity condition first, stimulus set A; and (4) low-creativity condition first, stimulus set B. The subject matter of the stimuli were the same across conditions for each version of the task, pertaining to such subjects as ecology, psychology, art history, and economics. See Appendix B for a complete list of the prompts.

The low-creativity condition included questions that could be answered directly by information found on Wikipedia pages (for instructions see Appendix A). For example, here is our ecology-related prompt: *How do species survive exceptionally harsh conditions?* While answering this question, participants visited Wikipedia pages about harsh environments and the animals that inhabit them. For instance, participants may start by exploring how animals survive in extremely cold environments, visiting pages about Antarctica or specific animals such as Emperor Penguins and Arctic Foxes (see Fig. 1). Subsequently, they might shift their focus to how animals adapt to extreme heat, searching Wikipedia pages about deserts or animals like Camels and Meerkats that thrive in such conditions (see Fig. 1). Finally, participants might investigate survival strategies in other extreme environments, such as the deep sea, by visiting pages about species like the Anglerfish (see Fig. 1). How animals survive harsh conditions has been heavily documented, and it is easy to find relevant information on Wikipedia pages. Thus, answering this question required low creativity.

The high-creativity condition includes questions that have yet to be answered by experts and/or lack a readily accessible, conclusive answer on Wikipedia. Instead, to answer the questions, participants had to generate a new answer that incorporated some of the information they found on Wikipedia. The questions in the high-creativity condition were intentionally ill-defined, whereas those in the low-creativity condition had well-defined answers. This distinction aligns with how creativity is often conceptualized: as the production of solutions to ill-defined problems (DeYoung et al., 2008; Finke et al., 1992; Mumford et al., 2012).

To emphasize that the answers were difficult to find on Wikipedia, participants were told that “The following questions cannot be fully answered by information found on Wikipedia. Instead, you will need to combine or synthesize information found while searching Wikipedia to construct an answer”. This directive was included to help participants manage the 15-minute time limit for each question. In more naturalistic settings, participants would discover for themselves that the information was not explicitly available. Although we acknowledge that this instruction may have shaped participants’ expectations, we believe it did not substantially alter their search behavior. Participants were still required to decide where to explore, when to switch, and how to generate responses, meaning that the fundamental explore-exploit dynamics of the task remained intact. What did change was that they did not have a period in which they checked whether they could directly find the answer.

Here is an example ecology-related stimuli for the high-creativity condition: *What are the evolutionary consequences of species becoming increasingly connected through globalization?* Unlike the low-creativity example, experts suggest that this is an open question (Sutherland et al., 2013). Participants often visited pages that related to specific topics mentioned in the prompt. For example, it was common for participants to visit pages on Globalization, Natural Selection, and Evolutionary

Biology. Participants were also seen to broaden their search by exploring more general pages about Biology and Global Biodiversity, or look into specific consequences such as Deforestation, Wet Markets, and Climate Change. Unlike the low-creativity question, participants may find themselves visiting more disparate pages or patches of ideas to come up with a solution.

To confirm that the participants understood the instructions for each condition, after reading the instructions they were asked “Can the following questions be fully answered by information found on Wikipedia?” The response options were yes, no, or maybe. If participants responded with “maybe”, their answer was not considered correct. We used these questions to exclude participants who did not comprehend the task instructions (i.e., by responding with the wrong answer both times they answered the question).

2.4. Knowledge-network analysis

To measure exploration and exploitation, we created knowledge interaction networks of the foraged-for information that capture the affinity of participants’ search behavior for focused, tightly interconnected knowledge patches or diverse loosely connected knowledge patches. To distinguish between phases of exploration and exploitation, we associated exploration with the looseness of a knowledge network and exploitation with the tightness of a knowledge network, using methods similar to Lydon-Staley et al. (2021). To measure looseness or tightness of a network, we created two types of knowledge networks that separated the tightness of knowledge between steps in a search trajectory (i.e., *sequential networks*) from the tightness of knowledge within the patches visited (i.e., *condition networks*). See Appendix C for the steps taken to preprocess Wikipedia pages prior to analysis.

First, we examined the sequential pages a person visited as they foraged for information, which we refer to as *sequential networks*. To do this, we created knowledge networks with Wikipedia pages as nodes, and connected pairs of pages if the participant visited the Wikipedia pages sequentially. Each participant had one network per stimulus, for a total of 4 networks (two for questions asked in the high-creativity condition and two for questions asked in the low-creativity condition). The edges were weighted by the semantic similarity between pairs of Wikipedia pages, by finding the cosine similarity of term-frequency inverse-document frequency (tf-idf) vectors of the text from those pages (see Appendix E for more information).

To measure the network structure, we calculated the average edge weight, which indicates the average semantic similarity or relatedness between the Wikipedia pages traversed by participants. If the average edge weight is lower, then this suggests that the participant is exhibiting more exploratory behavior by jumping around to relatively less similar pages. Alternatively, if the average edge weight is higher, then this can be perceived as more exploitative by visiting semantically similar content potentially from the same patch.

Next, we measured exploration and exploitation using a slightly modified version of the knowledge networks, which we refer to as *condition networks*. For the condition networks, all Wikipedia pages (i.e., nodes) were linked to all other Wikipedia pages visited by the participant in the same condition. As in the sequential networks, the edges were weighted by the cosine similarity of the tf-idf vector pairs of the text from the pages. We created two networks for each participant (one network for the high-creativity condition and one network for the low-creativity condition). Creating a fully connected network allowed us to measure a key characteristic of the tightness of a network, namely the clustering coefficient. The clustering coefficient of a graph is an indicator of how well a node is semantically connected to or clustered with other nodes in a network (Watts & Strogatz, 1998). This measure is calculated using triplets of connected nodes, which were rare in the sequential networks because it was uncommon for a participant

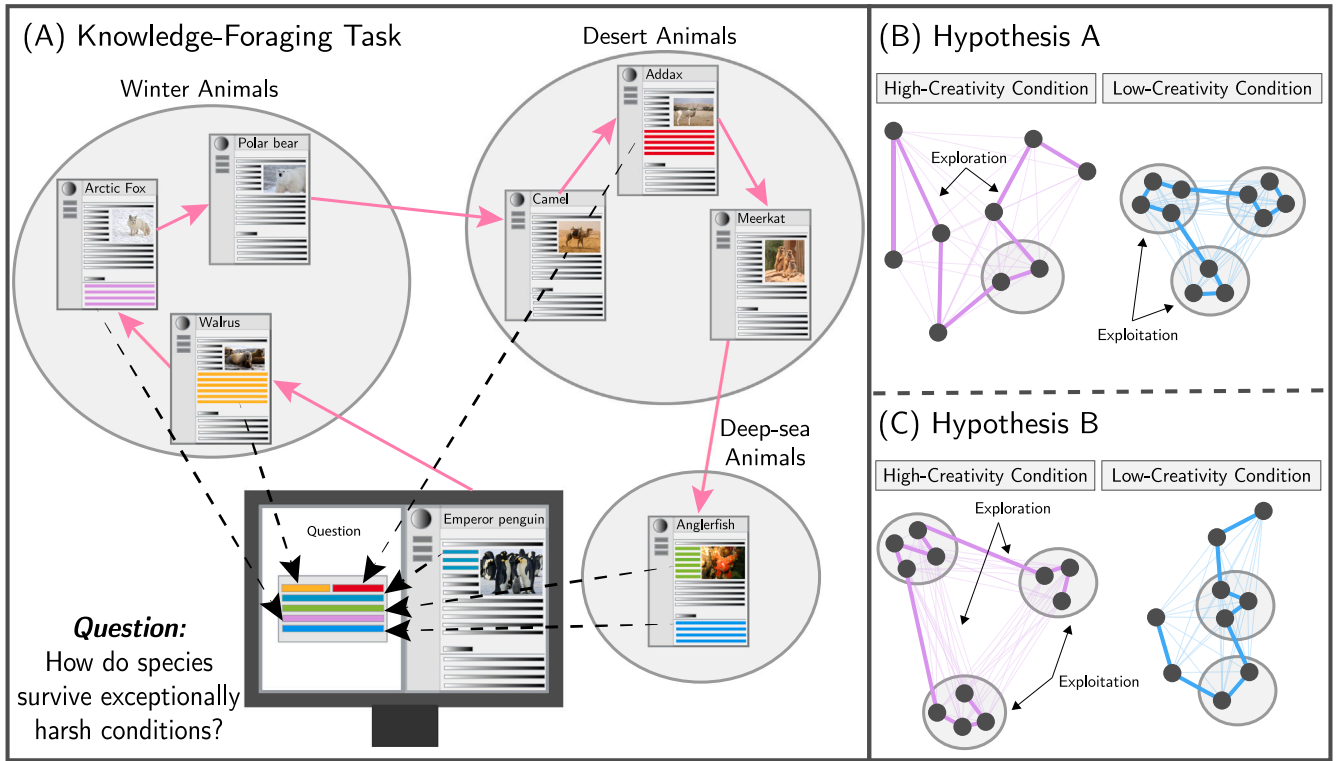


Fig. 1. Knowledge-Foraging Task. (A) Schematic illustration of the Knowledge-Foraging Task, where an example participant is asked to answer the following question: *How do species survive exceptionally harsh conditions?* The pink arrows indicate the order in which the participant visited the Wikipedia pages. The dotted-black arrows indicate that the participant incorporated information from the Wikipedia page into their answer. The information incorporated is the same color as the text from the Wikipedia page. Clusters of semantically similar information are enclosed in a circle. As you can see, the participant visited three distinct clusters or patches: one with articles about winter animals, one with articles about desert animals, and one with an article on deep-sea animals. (B) Illustration of Hypothesis A: high-creativity condition is associated with exploration (lower average edge weights, longer average path lengths, and lower average clustering coefficients), whereas the low-creativity condition is associated with exploitation (higher average edge weights, shorter average path lengths, and higher average clustering coefficients). The nodes represent Wikipedia pages visited by the participant. The edges are linked in two ways corresponding to our analysis: (1) a thick line that represents the sequential order in which the nodes were visited or (2) a thin line linking all nodes visited in the same condition. The semantic similarity is indicated by the distance between nodes, wherein nodes that are closer together are semantically similar. The gray circles illustrate nodes that are semantically similar and highly clustered. (C) Illustration of Hypothesis B: high-creativity condition is associated with both exploration (lower average edge weights, longer average path lengths) and exploitation (higher average clustering coefficients). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

to return to previously visited Wikipedia pages and thus form a fully-connected triplet. We used NetworkX to compute the average clustering coefficient C for each network (Hagberg et al., 2008):

$$C = \frac{1}{n} \sum_{i \in G} C_i$$

Where n is the number of nodes in graph G . C_i is the local clustering coefficient for node i , and is calculated by finding the geometric mean of the edge weights of a subgraph. The formula is as follows:

$$C_i = \frac{1}{deg(i)(deg(i) - 1)} \sum_{jk} (\hat{w}_{ij} \hat{w}_{ik} \hat{w}_{jk})^{1/3}$$

Where $deg(i)$ represents the degree for node i . j and k represent additional nodes in the network that form a triangle with node i . The connections of the triangle are weighted by w_{ij} , w_{jk} , and w_{ki} . The weights have been normalized by the maximum edge weight in the network (e.g., $\hat{w}_{ij} = w_{ij}/max(w)$). We can think of higher clustering as indicative of exploitation, wherein a person is visiting highly related Wikipedia pages that form clusters or patches.

We also measured the average shortest path length, which is an indicator of the average shortest distance between all pairs of nodes in a network (Watts & Strogatz, 1998). The shortest path between two nodes is the one where the sum of the edge weights traversed produces the smallest value. The average shortest path length indicates how

easy it is to traverse a network. In our networks, a higher value of semantic similarity between two nodes indicates that the two nodes are closer together in semantic space, and thus have a smaller distance. To transform our edge weights so that lower values indicate closer proximity, we inverted our semantic similarity measure (i.e., $1 - \text{semantic similarity}$). We used the average shortest path length function from NetworkX to calculate the average shortest path length P (Hagberg et al., 2008):

$$P = \frac{1}{n(n-1)} \sum_{s,t \in V, s \neq t} d(s,t)$$

Where n is the number of nodes in the network and V is the set of nodes. $d(s,t)$ represents the shortest weighted path length between nodes s and t .

A shorter average path length is indicative of a tighter knowledge network, where the Wikipedia pages are more interconnected. A longer average path length is indicative of a looser network that takes longer to traverse because the Wikipedia pages are semantically dissimilar. We can think of a longer average path length as representative of exploration, or as taking more or larger steps between pages or patches. Alternatively, shorter average path length may indicate exploitation in that the path between pages is relatively short, suggesting that the pages are comparatively more similar.

3. Results

3.1. People trade-off between exploration and exploitation during the creative task

We found evidence that people trade-off between exploration and exploitation when asked to create an original answer (see Fig. 2). First, we used our sequential networks (where nodes were connected if they were sequentially visited by the participant) to test how the average edge weight differed between the conditions. We found that the average edge weights (or similarity between Wikipedia pages) were significantly smaller for the high-creativity condition ($M = 0.14$) than for the low-creativity condition ($M = 0.17$; $t = -3.41$; $p < 0.001$; Cohen's $d = -0.49$). To address the concern that averages may be sensitive to heavy-tailed distributions, we repeated our analysis using the median edge weights in addition to the mean. This additional measure pointed to a similar conclusion: the median edge weights were significantly smaller in the high-creativity condition ($M = 0.14$) than in the low-creativity condition ($M = 0.17$; $t = -3.19$; $p = 0.002$; Cohen's $d = -0.47$). This consistency across mean and median measures indicates that our findings are robust and not driven by outliers or heavy-tailed distributions. This suggests that coming up with a highly creative answer involved jumping from relatively more semantically dissimilar pages, forming looser knowledge networks of foraged-for information.

Next, we ran analyses using the condition networks, where nodes (i.e., Wikipedia pages) are connected if they are visited by the participant in the same condition. We were unable to compare the average shortest path length and the average clustering coefficient between conditions using a paired t-test because our data violated the normality assumption (Shapiro–Wilk normality test: $p < 0.05$); therefore, we used the Wilcoxon two-sample paired signed rank test (Wilcoxon, 1945). We found that the average shortest path length did not differ between the high-creativity condition ($Mdn = .90$) and the low-creativity condition ($Mdn = 0.91$; $V = 2,084$; $p = 0.17$). However, we found that the average clustering coefficient was significantly larger for the high-creativity condition ($Mnd = 0.32$) than for the low-creativity condition ($Mnd = 0.23$; $V = 2,734$; $p < 0.001$; Effect size $r = 0.53$). This suggests that when creating original responses, participants are exploiting semantically similar information. These findings replicate findings from our pilot study with 64 participants, where we found that the condition networks in the high-creativity condition exhibited higher average clustering coefficients (see Appendix G for a discussion of the pilot study).

The higher clustering coefficients in the high-creativity networks may have influenced our measure of average shortest path length, contributing to the null finding. Because many concepts formed tightly interconnected triads, short paths occurred between nodes. These connections can offset longer edges and keep overall path lengths comparable across conditions. Consistent with this interpretation, within-subject increases in clustering were negatively associated with average path length (Spearman's $\rho = -0.44$), suggesting that higher clustering systematically compressed path length values. By also analyzing sequential networks, we were able to detect differences in exploratory behavior that would otherwise have been masked by the high clustering of the condition networks.

Together, these findings suggest that creative foraging involves trading-off between exploration and exploitation. We can interpret the participants' behavior as exploiting patches of semantically similar information before exploring a semantically distant patch or page. In other words, taking large leaps between semantically dissimilar patches.

Furthermore, these differences are unlikely to be attributed (or only attributed) to the number of pages visited or the methods by which participants navigated to them. The conditions were not significantly different in terms of the number of pages visited per stimulus (high-creativity: $M = 4.40$; low-creativity: $M = 4.66$; $t = -0.97$; $p = 0.34$), the

average number of times they navigated to a new page by clicking a hyperlink (high-creativity: $M = 2.48$; low-creativity: $M = 2.83$; $t = -1.64$; $p = 0.10$), and the average number of times participants searched a term to find a page (high-creativity: $M = 1.62$; low-creativity: $M = 1.34$; $t = 1.84$; $p = 0.07$). To further assess the marginal effects, we calculated Bayes Factors. The Bayes Factors for hyperlink clicks ($BF_{10} = 0.37$) and the number of times participants searched a term ($BF_{10} = 0.51$) provided anecdotal to moderate evidence in favor of the null, suggesting that participants in the two conditions did not reliably differ in these behaviors.

3.2. People provide more original answers during the creative task

To test whether our manipulation was successful — in that the answers from the high-creativity condition were indeed more original when compared to the answers from the low-creativity condition — we ran a series of semantic similarity analyses. If our manipulation was successful, we expected to see lower semantic similarity between the answers from the high-creativity condition and the text from the Wikipedia pages. Before running these analyses, we excluded answers that were deemed off-topic or irrelevant to the question to ensure that differences between conditions were not simply due to meaningless deviations from the text (e.g., greater semantic distance in the high-creativity condition because of a higher number of irrelevant responses). To do this, we used GPT-5 to classify each answer as on-topic or off-topic in relation to the question (See Appendix F for more information). To validate GPT's ratings, a human rater independently judged 20% of the responses (97 answers). Agreement between GPT and the human rater was substantial (Cohen's Kappa = 0.80; percent agreement = 97.90%). Given this level of agreement, we relied on GPT's ratings for the remaining responses in our analyses. GPT identified 26 instances where participants' answers were off-topic, which were subsequently excluded from the analyses. Although we excluded responses that were clearly off-topic, semantically distant responses could still be inaccurate or misleading. Accuracy could, in principle, be evaluated in the low-creativity condition, where correct answers are more clearly defined, but this metric does not generalize to the high-creativity condition where there is not a single correct answer. For this reason, we did not include an accuracy analysis in the present study. See Appendix D for the steps taken to preprocess the participants' responses prior to analysis.

First, we started by comparing the semantic similarity between the participants' answers and the text from each Wikipedia page, by finding the cosine similarity between tf-idf vector pairs (see Appendix E). We were unable to compare the conditions using a paired t-test because our data violated the normality assumption (Shapiro–Wilk normality test: $W = 0.97$; $p = 0.02$); therefore, we used the Wilcoxon two-sample paired signed rank test (Wilcoxon, 1945). The average semantic similarity between the participants' answers and text from the Wikipedia pages was significantly lower for the high-creativity condition ($Mdn = 0.04$) than for the low-creativity condition ($Mdn = 0.06$; $V = 2,492$; $p = 0.01$; Effect size $r = -0.27$; see Fig. 3). This result indicates that participants in the high-creativity condition were providing more original answers, that relied less on the foraged-for text. This result also substantiates the intended claim that participants were more creative in the high-creativity condition and less creative in the low-creativity condition. Note that these results are unlikely to be attributed to differences in answer length, as the average number of words in the answers did not differ significantly between conditions ($t = -1.81$; $p = 0.07$; $BF_{10} = 0.50$).

We also found that the meaning of individual sentences from the participants' answers differed more from the meaning of the sentences from the Wikipedia pages they visited (see Appendix E for calculation using sentence transformers). We were unable to compare the conditions using a paired t-test because our data violated the normality assumption (Shapiro–Wilk normality test: $W = 0.96$; $p =$

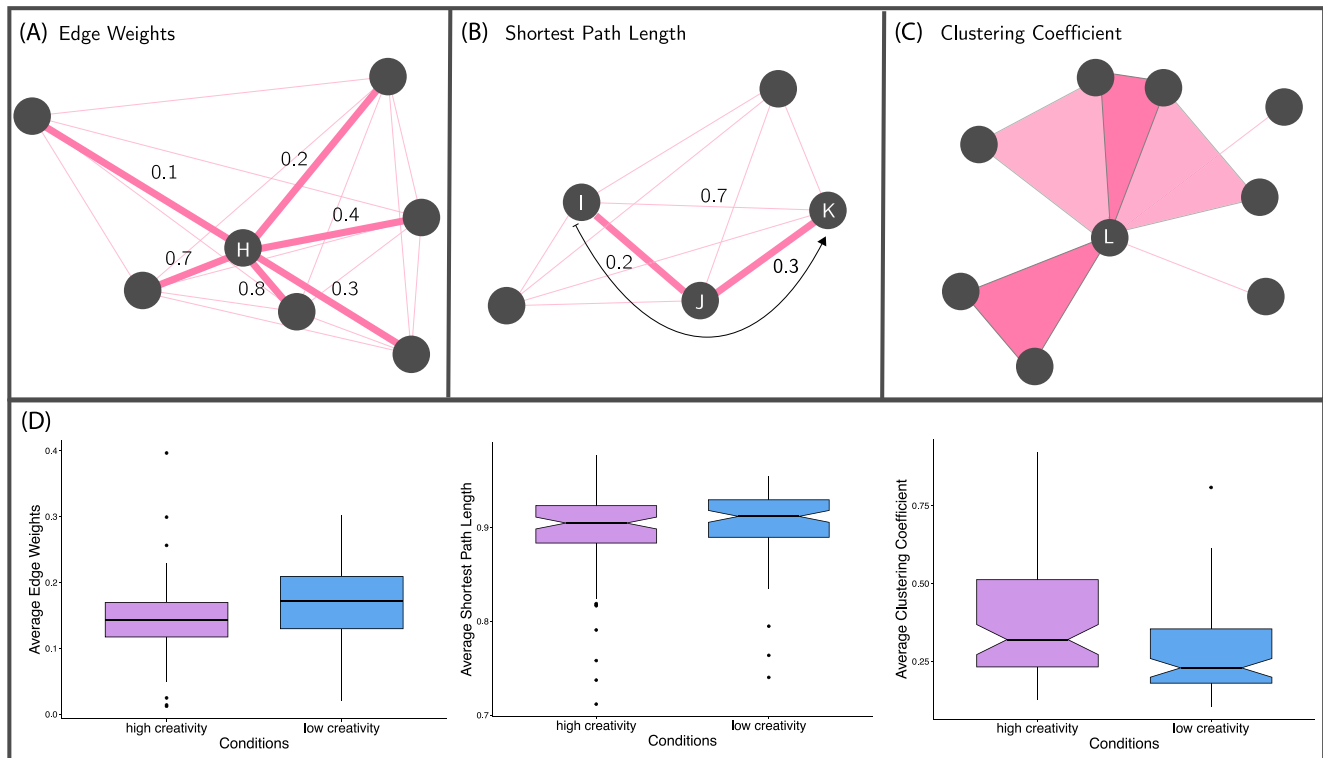


Fig. 2. Knowledge-Network Analysis. (A) Illustration of edge weights, wherein the edge weights for node *H* are represented. Higher edge weights indicate that the pair of nodes are more semantically similar. (B) Illustration of shortest path length, where the shortest path from *I* to *K* is through *J*. The shortest path is the path where the edge weights sum to the lowest number. We inverted our semantic similarity measure so that lower numbers were indicative of higher similarity (i.e., $1 - \text{semantic similarity}$). Although *I* and *K* are directly connected, the edge weight of this path (i.e., 0.7) is larger than the sum of the edge weights for the path through *J* (i.e., 0.5). (C) Illustration of clustering coefficient, where the clustering coefficient for *L* is the extent to which its neighbors are connected (measured by calculating the geometric mean of all triangles in which *L* belongs). (D) Box plots of the results showing that the knowledge networks for the high-creativity condition — when compared to the low creativity condition — exhibited lower average edge weights (left-most plot) and higher average clustering coefficients (right-most plot). The conditions were not significantly different in terms of the average shortest path lengths (middle plot).

0.001); therefore, we used the Wilcoxon two-sample paired signed rank test (Wilcoxon, 1945). Specifically, we found that the average semantic similarity between each sentence of the participants' answer and the best matching sentence from the Wikipedia pages was significantly lower for the high-creativity condition ($Mdn = 0.66$) than for the low-creativity condition ($Mdn = 0.72$; $V = 1,413$; $p < 0.001$; Effect size $r = -0.58$; see Fig. 3). We therefore find additional support that the answers from the high-creativity condition deviated more from the Wikipedia text than the low-creativity condition.

Finally, we tested how well the topics mentioned in the participants' answers mapped onto the topics mentioned in the Wikipedia pages. We expected that the topics mentioned in the high-creativity condition would be comparatively less similar to the topics mentioned in the Wikipedia pages; whereas the topics mentioned in the low-creativity condition would be relatively more similar to the topics mentioned in the Wikipedia pages. To model the topics, we used the Latent Dirichlet Allocation (LDA) topic modeling technique (see Appendix E and F). We compared the distribution of topics by calculating the correlation between the distribution of topics associated with the participants' answers and the distribution of topics associated with each Wikipedia page visited by the participant.

We found that the average correlation between the topics found in the participants' answers and the topics represented in the Wikipedia pages were significantly lower for the high-creativity condition ($M = 0.26$) than for the low-creativity condition ($M = 0.54$; $t = -10.59$; $p < 0.001$, Cohen's $d = -1.13$; see Fig. 3). This result suggests that the high-creativity condition prompted participants to write about topics distinct from the ones they read on Wikipedia pages, providing additional

evidence that the answers in the high-creativity condition were more original than the ones in the low-creativity condition.

Overall, the analyses in this section suggest that the text from the high-creativity condition (when compared to the low-creativity condition) was significantly less similar to the text on the Wikipedia pages. That is, participants in the high-creativity condition were more likely to produce original or creative answers. In line with this finding, we also see that participants were less likely to paste text from the Wikipedia pages into the textbox on average in the high-creativity condition ($M = 1.52$) than those in the low-creativity condition ($M = 2.15$; $t = -3.80$; $p < 0.001$). These results provide additional evidence that the informational context differed between conditions: it was harder to find the answers in the high-creativity condition than in the low-creativity condition.

3.3. Individual difference in originality associated with exploration

In addition to comparing across conditions, we also examined how individual differences in originality related to the structure of the knowledge networks participants built. We quantified originality by measuring the average semantic similarity between each response and all other responses to the same stimulus. Each response was embedded using the all-MiniLM-L6-v2 sentence transformer model. This model is well-suited for capturing semantic similarity between short responses, allowing us to generate vector representations of their meaning (Reimers & Gurevych, 2019). We began by computing all pairwise cosine similarities between embeddings within the same stimulus and condition. Then, for a given response i , we calculated its mean similarity to all other responses $j \neq i$, and used this to calculate the originality

score: $\text{originality}_i = 1 - \text{mean similarity}_i$. Higher values indicate greater semantic distance from other responses to the same stimulus. We then averaged these values across all responses produced by a participant in a condition to obtain a condition-level originality measure. In line with our findings from Section 3.2, the average originality score was significantly higher for the high-creativity condition ($Mdn = 0.39$) than for the low-creativity condition ($Mdn = 0.36$; $V = 2,994$; $p < 0.001$; Effect size $r = 0.46$).

We then examined how these originality scores related to exploration and exploitation. In linear mixed-effects models, with subject as a random intercept and our network measures as predictors, we consistently found that edge weights were reliable negative predictors of originality. We ran two separate models: one that included average edge weights and another that included median edge weights. When including average edge weight as a predictor, lower values predicted higher originality ($\beta = -0.019$, $SE = 0.007$, $p = .007$). Neither average shortest path length ($\beta = -0.013$, $SE = 0.007$, $p = .069$) nor average clustering coefficient ($\beta = -0.003$, $SE = 0.005$, $p = .547$) were significant predictors. When including median edge weight instead, we observed a similar negative relationship with originality ($\beta = -0.019$, $SE = 0.007$, $p = .005$), while again neither average shortest path length ($\beta = -0.013$, $SE = 0.007$, $p = .059$) nor average clustering coefficient ($\beta = -0.003$, $SE = 0.005$, $p = .627$) reliably predicted originality.

Taken together, these results suggest that, when considering originality relative to other answers to the same stimuli, individual differences emerged. Participants who generated more original answers tended to explore semantically dissimilar pages.

3.4. Marginal value theorem not predictive of foraging behavior

We also investigated whether the behavior exhibited by participants while doing the Knowledge-Foraging Task aligns with optimal foraging. Optimal foraging theory posits that organisms seek to maximize the acquisition of resources (e.g., food or information) while minimizing associated costs such as time and effort. One key optimal foraging model is Marginal Value Theorem (MVT), which posits that an optimal forager should leave a patch when the marginal rate of resource acquisition falls to match the average rate of acquisition in the environment (Charnov, 1976). Previous literature suggests that optimal foraging may be more applicable to non-creative contexts. For instance, Hart et al. (2017) found that people deviate from the optimal foraging theory when tasked with creatively generating novel shape configurations. We wanted to extend these findings to examine how this relationship unfolds in an informational context, where participants either generate a creative answer or search for an existing answer.

We hypothesized that participants in low-creativity contexts would be more likely to adhere to the Marginal Value Theorem, as the decision criteria for resource allocation are more straightforward. Specifically, when participants can locate information verbatim on a page, they can easily gauge whether that page continues to yield relevant content; thus, the rate at which they encounter useful information is a clear indicator of whether to stay or move on. In high-creativity contexts, the frequency of partially matching information on a page is an unreliable predictor of whether participants will ultimately find the relevant information needed to complete their answer. Whether the information rate on a page is high or low, the preceding rate of partially matching information on a page is an unreliable guide to whether they will eventually discover the exact information needed to complete their answer. Search failure can follow high rates of strongly-related information, while a key fact may be encountered on a page with only tangentially relevant content. In these circumstances, MVT predictions should break down as there is no incremental information to gather, and determining when to leave cannot be as easily based on the rate at which information is found.

To test this, we examined how well each condition adhered to the MVT predictions. In this analysis, each Wikipedia page represented

a “patch” from which participants gathered information for their answers. This differs from our network analyses, wherein patches are considered to be highly clustered Wikipedia pages. A “piece of information” was considered to be collected when a participant navigated from the patch (Wikipedia page) back to the textbox, as indicated by the cursor entering the textbox area. This action suggested that the participant was inspired by something encountered on the corresponding Wikipedia page and decided to update their answer.

We aimed to evaluate how well the MVT predicted participants’ decision to leave a Wikipedia page in search of a new one. To assess adherence to the MVT, we calculated the rate of information acquisition for each page visited. This was based on the rate at which they gathered pieces of information on the current page, as indicated by moving their cursor to the textbox from the Wikipedia side of the screen. This rate was determined by dividing the number of “pieces of information” collected on a page, $n(t_i^{\text{info}})$, by the total time spent, which included both the time on the page up until the last piece of information was collected, t_i^{info} , and the average travel time between pages, T :

$$\text{Rate of information acquisition} = \frac{n(t_i^{\text{info}})}{t_i^{\text{info}} + T}$$

Travel time, T , was defined as the time it took to find a viable page where the participant collected a piece of information. Travel time before arriving at the first page tended to be slightly longer, as it included the time participants spent reading the new prompt.

According to the MVT, a participant should leave the current page when the marginal rate of information acquisition equals the average rate of acquisition. We approximated the marginal rate of information acquisition as the reciprocal of the time each participant spent on a page before transitioning to a new page after collecting their final piece of information, denoted as:

$$\text{Marginal rate of information acquisition} = \frac{1}{t_i^{\text{exit}}}$$

This value represents an estimate of the marginal rate of information acquisition at the point participants decided to leave a page, which we operationalized as the inverse of the time elapsed between collecting the last piece of information and exiting the page. This operationalization follows the definition of the marginal rate of gain at the decision point in the classical Marginal Value Theorem (Charnov, 1976). By evaluating the rate at the moment of departure, rather than incorporating subsequent navigation or search on the next page, this definition aligns with the theoretical basis of optimal foraging and with common practice in the literature. The last page participants visited before moving on to the next stimulus was excluded from our calculations because we could not determine t_i^{exit} , the interval between incorporating the last piece of information and transitioning to a new page.

To evaluate how closely participants’ behavior aligned with the point at which the MVT predicts they should leave a page, we calculated the difference between the marginal rate of information acquisition and the average rate of information acquisition: $\frac{1}{t_i^{\text{exit}}} - \frac{n(t_i^{\text{info}})}{t_i^{\text{info}} + T}$. A value close to zero indicates strong alignment with the MVT, while larger values suggest deviations from the predicted behavior. To quantify the magnitude of deviation, regardless of whether $\frac{1}{t_i^{\text{exit}}}$ was greater or less than the rate of information acquisition, we calculated the absolute value of the difference:

$$\text{Absolute Difference} = \left| \frac{1}{t_i^{\text{exit}}} - \frac{n(t_i^{\text{info}})}{t_i^{\text{info}} + T} \right|$$

We used the Absolute Difference to assess how well participants’ foraging behavior aligned with the MVT under each condition. After excluding participants for whom we were unable to calculate the Absolute Difference in both conditions, often due to missing recordings of relevant cursor movements, the final sample size was reduced to 117

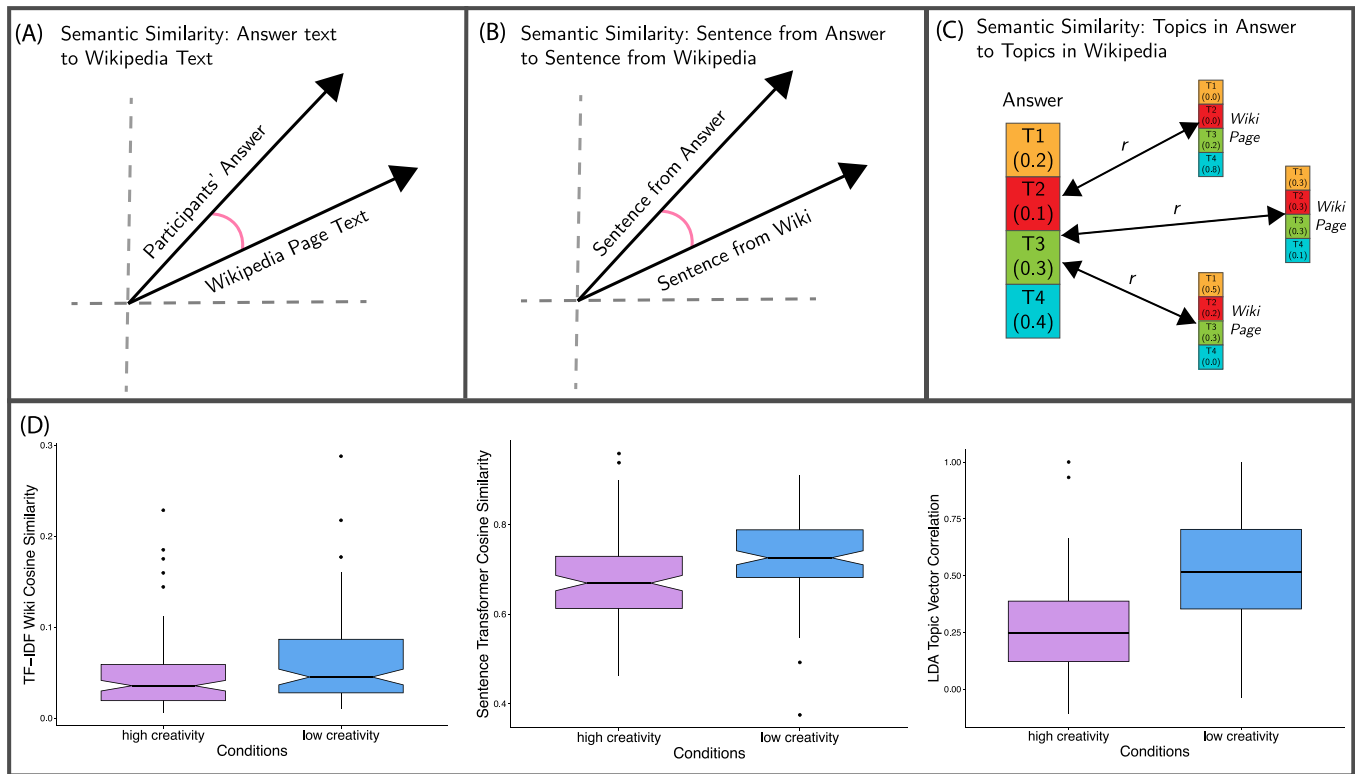


Fig. 3. Originality of Answers Analysis. (A) Illustration of semantic similarity between a participant's answer (represented as a tf-idf vector) and one of the Wikipedia pages they visited (represented as a tf-idf vector). The semantic similarity score is calculated based on the cosine similarity between the two tf-idf vectors. (B) Illustration of the semantic similarity between a sentence from the participant's answer (represented as a sentence embedding) and one of the sentences from one of the Wikipedia pages they visited (represented as a sentence embedding). The semantic similarity score is calculated based on the cosine similarity between the two sentence embeddings. (C) Illustration of the similarity between the topic distribution for a participant's answer and the topic distributions for the Wikipedia pages they visited. The similarity between the topic distributions is measured by calculating the correlation between the pairs of distributions. (D) Box plots of the results showing that the high-creativity condition — when compared to the low creativity condition — exhibited lower semantic similarity between the text from the participants' answers and the text from the Wikipedia pages (left), lower semantic similarity between the sentences from the participants' answers and the sentences from the Wikipedia pages (middle), and lower average correlations between the answers' topic distributions and the Wikipedia pages' topic distributions (right).

participants. Due to the non-normal distribution of these differences (Shapiro–Wilk test, $p < 0.001$), we conducted a Wilcoxon signed-rank test, which revealed no significant difference between the two conditions (Wilcoxon signed-rank test = 1,945.0, $p = 0.357$). These results suggest that participants' adherence to the MVT was similar in both the low-creativity and high-creativity conditions.

Next, we examined whether participants' observed patch-leaving rates deviated significantly from the predictions of the MVT in both conditions. Specifically, we tested whether the difference between the marginal rate of information acquisition and the average rate of information acquisition (i.e., $\frac{1}{i_{\text{exit}}} - \frac{n(i_{\text{info}})}{i_{\text{info}} + T}$) significantly differed from 0. A value of 0 would indicate exact adherence to the MVT. Unlike when we calculated the Absolute Difference (above), for this calculation we did not convert the difference to its absolute value. For the low-creativity condition, the Wilcoxon signed-rank test showed a significant difference from MVT predictions ($W = 617.0$, $p < 0.001$), with a large effect size ($r = 0.696$). In the high-creativity condition, the Wilcoxon signed-rank test again indicated a significant difference from MVT predictions ($W = 342.0$, $p < 0.001$), with a large effect size ($r = 0.748$). Combining both conditions, the Wilcoxon signed-rank test revealed a significant difference from MVT predictions ($W = 1,859.0$, $p < 0.001$), with a large effect size ($r = 0.722$). These results indicate that participants' foraging behaviors deviated significantly from MVT predictions in both conditions.

4. Discussion

In this study, we used our newly developed Knowledge-Foraging Task to measure differences in foraging behavior based on the level of creativity required to complete the task. We manipulated the level of creativity by having participants answer questions that were either easy or difficult to answer with information on Wikipedia pages, and tested how creativity impacted exploration and exploitation. When the information was difficult to find, participants had to come up with original or novel answers (high-creativity condition). Alternatively, when the information was easy to find, participants could directly use it to formulate an answer (low-creativity condition). Our results show that people trade-off between exploration and exploitation when tasked with generating a creative answer. In particular, we found that the knowledge networks in the high-creativity condition exhibited more exploration (as indicated by lower average edge weights) and more exploitation (as indicated by higher average clustering coefficients).

Foraging is often associated with striking a balance between exploration and exploitation (Hart et al., 2017; Hills et al., 2008, 2015), and this balance has recently been linked to creative foraging in a perceptual context (Hart et al., 2022, 2017). Trading off between exploration and exploitation has also coincided with improvements in problem solving (Uotila et al., 2009) and reinforcement learning (Cohen et al., 2007; Hills et al., 2015; Sutton & Barto, 1998). Our work extends the existing literature to a naturalistic information-seeking context,

providing valuable insights into how people search for information when it is readily available versus when it requires deeper, creative thought. Observing that the exploration–exploitation trade-off emerges when people generate creative answers offers a unique perspective on the dynamic processes behind information search, emphasizing the significance of contexts that foster both breadth and depth in search strategies. These findings have broader implications for understanding how creative processes unfold and for designing environments that foster innovation and discovery.

One possible interpretation of these findings is that creative foraging involves exploiting diverse patches of Wikipedia pages. The lower average edge weights may be caused by a larger semantic distance between patches of Wikipedia pages. Therefore, participants can be seen as jumping from diverse clusters of tightly connected ideas as they build their responses. The general notion of jumping between disparate patches of information maps well to descriptions of exploration and exploitation in a foraging context (Hart et al., 2022, 2017; Hills et al., 2015), providing additional evidence for this explanation.

Visiting disparate patches in the high-creativity condition may reflect the consideration of distinct hypotheses. That is, participants might investigate a set of related Wikipedia pages centered around a specific hypothesis to determine if it addresses the question. They may extract only the most salient information from each patch—a behavior that conceptually overlaps with scrounging when socially embedded linguistic cues suggest that certain information is “better” than other items within the same patch (Giraldeau & Caraco, 2000). If the hypothesis proves unfruitful, they might revisit the question, consider its relationship to a different hypothesis, and pursue a new set of Wikipedia pages. This shift in focus could account for the longer leaps observed between clusters or jumps from clusters to distinct pages. Alternatively, the exploration–exploitation trade-off may arise from participants comparing and integrating information from other pages, both within and across clusters. Comparing information within a cluster could involve reading about related concepts to develop a more comprehensive understanding, engaging in a “deep-dive” into a specific area before transitioning to another patch. In such cases, they may follow an expanding specialist strategy, starting with a narrow focus and gradually broadening the scope of inquiry. Comparing information across clusters or concepts may involve integrating or synthesizing disparate information. Developing a deep understanding of distinct ideas and potentially recombining or synthesizing them could foster the emergence of new ideas, as required in the high-creativity condition.

In contrast, participants in the low-creativity condition may visit pages that are already closely connected, sharing similar content on related topics. This overlap could result in participants searching less distinct clusters. Additionally, the participants may focus primarily on comparing the information they find directly to the prompt to determine if it answers the question. This approach may lead to behavior that is less indicative of deep and broad searches, and more linear in nature.

Additionally, our novel paradigm enabled us to capture not only click-through data (e.g., hyperlink clicks) but also the text participants generated based on the content of the Wikipedia pages they visited. While click-through data alone offers limited insight into what participants take away from each page, our paradigm provided a more detailed understanding of the specific information they gathered and utilized. We found that the content of the answers from the high-creativity condition were more original when compared to the answers from the low-creativity condition, as indicated by a greater deviation from the text on the Wikipedia pages participants visited. Specifically, while comparing answers from the high-creativity condition to those from the low-creativity condition, we found: (1) the participants’ answers were less semantically similar to the entire text of the Wikipedia pages; (2) each sentence in the participants’ answer showed less semantic similarity to the sentences in the Wikipedia pages; and (3) the topic probability distribution linked to the participants’

answers was less similar to the topic probability distribution associated with the Wikipedia pages. These results link the development of novel answers to information-seeking behavior that exhibits the exploration–exploitation trade-off. It also suggests that our manipulation was successful: that people provided more original or novel answers in the high-creativity condition.

This difference in originality could arise for multiple reasons. In the high-creativity condition, participants not only exploit closely related information by visiting pages on connected concepts but also explore distinct topics. As they synthesize this information, their responses may diverge significantly from the text they initially read, reflecting a higher level of originality. Participants could also be incorporating more information from their prior knowledge into their answers. Both options involve the reconfiguration or manipulation of information to derive a new answer, which is often referred to as sense-making (Pirolli, 2005). Further research can probe what contributes to the originality of the creative answers by asking for more clarification from participants as to how they answered the questions.

Note that exploratory search behavior does not necessarily equate to creativity in the response. For example, one could design a question in the low-creativity condition where the required information is explicitly available on Wikipedia pages but still necessitates sampling from multiple, highly distinct pages (e.g., “How many times does the word creativity appear in pages related to Guns, Germs, and Steel?”). In such cases, the knowledge network may reflect exploration through visits to distinct pages (i.e., lower average edge weights). However, the answer itself would be directly determined by information present on those pages and would not require participants to generate an original response that extends beyond the given content (e.g., the number of times creativity appears on those pages). This underscores that our operationalization of creative foraging hinges on participants producing responses that move beyond the direct information found on the pages.

Our study indicates that the Marginal Value Theorem (MVT) does not accurately predict human foraging behavior in the Knowledge-Foraging Task, as participants in both the low-creativity and high-creativity conditions showed significant deviations from MVT predictions. MVT may break down in environments where the predictability of information acquisition is low, such as in our task where participants may have engaged in non-incremental strategies (Davidson & Hady, 2019). For example, some participants may have chosen to read an entire Wikipedia page or section before assessing its relevance and updating their answer, rather than updating their answer after encountering each relevant piece of information. This suggests that the “taking” of information may occur after reviewing an entire page rather than incrementally, rendering MVT inapplicable in this context. Additionally, suboptimal decisions observed in our study, such as staying on a page longer than MVT would predict, may arise from the combination of environmental complexity and cognitive effort required to assess whether a Wikipedia page contained relevant information (Davidson & Hady, 2019; Wilke et al., 2009b). Alternatively, these findings could be influenced by methodological limitations, such as relying on cursor movements to approximate information acquisition, which may not accurately reflect participants’ thought processes or whether they utilized the information. Future research could more precisely determine whether participants collect or incorporate a “piece of information” into their answers by requiring them to explicitly indicate relevant information or by using think-aloud protocols.

Our study is not without limitations. One limitation was the number and diversity of the stimuli. Participants were only asked to provide answers to four of the eight stimuli (two of the four stimuli for the high-creativity condition and two of the four stimuli for the low-creativity condition). We prioritized the duration of time allotted to the stimuli, rather than the number of stimuli. During our pilot study, participants saw all eight stimuli, with 5 min to answer each question. We found that 62 out of our 64 participants visited less than three Wikipedia pages for one or more of the stimuli during the five minutes, rendering

their data obsolete for our clustering analysis. Therefore, in attempts to increase the number of pages visited by the participants, we increased the time to 15 min and reduced the number of stimuli. Although we increased the time to 15 min, the time could have been longer. To prevent mental exhaustion from influencing the participants' responses, we did not want the task to take longer than 1 h. We expect that, with more time, participants would be able to visit more Wikipedia pages, resulting in larger and richer knowledge networks. Also, the high-creativity stimuli differed from the low-creativity stimuli not only in being more ill-defined, but also in terms of breadth. Ill-defined questions are those that can be answered in many different ways, with no pre-specified correct answer. By contrast, broad questions abstract away from specific instances and instead integrate across them. For instance, the question "How is evolution impacted by globalization?" is broader in scope than the more specific question "How do species survive harsh conditions?". However, the breadth of the questions were not always broad in the high creativity condition. For example, the high-creativity question "What brain mechanisms help people make scientific discoveries?" could be answered quite specifically by citing particular neural mechanisms. Thus, while questions in the high-creativity condition are generally broader and tend to invite more abstract answers, not all high-creativity questions necessarily require broad or abstract answers. Future research could disentangle these dimensions by systematically manipulating the level of abstraction while holding ill-definedness constant (or vice versa), thereby clarifying their independent contributions.

Another limitation is our inability to fully confirm that the answers to the questions from the high-creativity condition were completely absent from Wikipedia. Our research team validated the conditions by attempting to find the information on Wikipedia. For the high-creativity condition, they searched relevant key terms in attempts to find the answers on Wikipedia, but to no avail. However, for the low-creativity condition, the answers to the questions were easy to access by searching terms mentioned in the prompts. We found further support for the ease with which information can be found through our semantic analysis, showing that answers from the high-creativity condition (when compared to the low-creativity condition) were less relevant to the text on the Wikipedia pages. These findings suggest that the information that they sought was less accessible.

A final limitation of our study lies in the inherent constraints of the paradigm. First, we could not track whether participants accessed external websites outside our cloned Wikipedia environment. This poses a potential concern, as outside information might have influenced their responses. However, we minimized this risk by explicitly instructing participants to rely solely on the provided materials and by incorporating validity checks to ensure they read and understood the instructions. Second, although we cloned Wikipedia's content at the time of extraction, the search page options in our version differed from those on the official site. Even with this difference, participants appeared to navigate the platform effectively. For instance, in the low-creativity condition, their responses aligned more closely with the available information, indicating that they accessed pages with more relevant content than those visited in the high-creativity condition. Third, our paradigm does not capture whether participants' responses were based on their prior knowledge rather than on information gathered from Wikipedia. Some questions could plausibly be answered using prior knowledge without integrating content from the Wikipedia pages they viewed. Our measure of originality captures only the degree of deviation from Wikipedia, not whether that deviation reflects creative synthesis or reliance on prior knowledge. Future work could address this issue by asking participants to self-report whether the information in their answers was inspired by what they read on Wikipedia or derived from their prior knowledge.

In summary, our research suggests that creativity influences the foraging strategies employed while searching for information. We tracked the information-seeking behavior of individuals as they scoured

Wikipedia for relevant facts, manipulating the ease with which they could find pertinent information. Our findings indicate that in situations of information scarcity, individuals produced original or creative answers by balancing periods of exploration and exploitation. These insights shed light on how learners and scientists forage for information while crafting new theories or explanations.

CRediT authorship contribution statement

Kara Kedrick: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kyana Burhite:** Writing – review & editing, Software, Methodology, Investigation, Formal analysis, Data curation. **Iris Vilarés:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Paul R. Schrater:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.cognition.2026.106451>.

Data availability

The data, along with the Python 3 and R code used to analyze and visualize it, are available at OSF: https://osf.io/n4sb8/overview?view_only=935781672123414ebd330afd8313821d.

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